MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

III SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
END SEMESTER EXAMINATIONS, NOVEMBR – DECEMBER 2024

ELECTRICAL CIRCUIT ANALYSIS [ELE 2121]

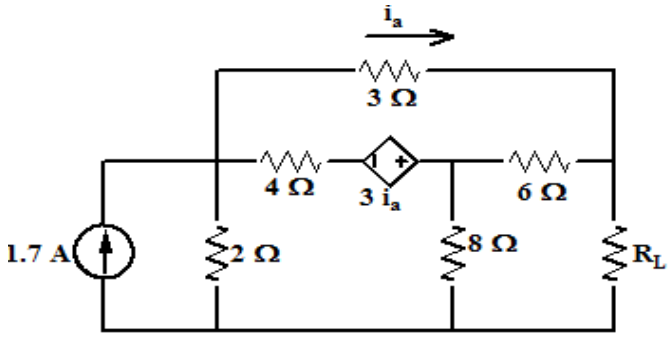
REVISED CREDIT SYSTEM

Time: 3 Hrs. **Date: 27 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

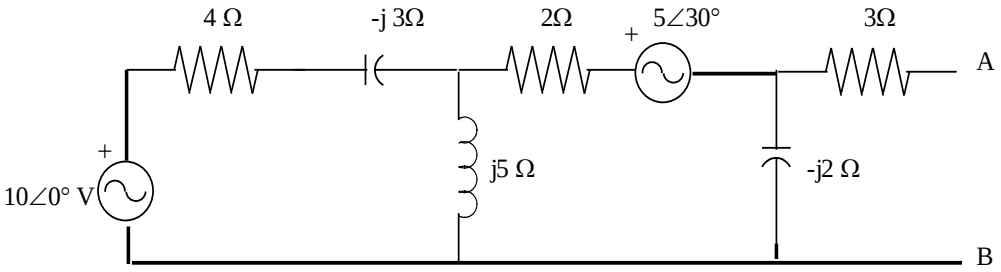
- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Obtain the Thevenin`s equivalent circuit across 'R_L ' for the circuit shown below.



(04)

1B. For the network shown, determine the load impedance to be connected across AB for maximum power transfer, if the load consists of a variable resistance and a variable reactance.



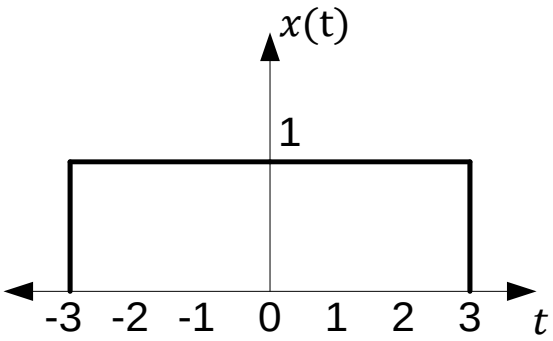
(03)

1C. Determine the Energy and Power of the signal given below. Also comment on the type of the signal based on the above analysis.

$$x(t) = \begin{cases} 2\left(1 - \left|\frac{t}{4}\right|\right) & ; for\ |t| \le 4 \\ 0 & ; otherwise \end{cases}$$

(03)

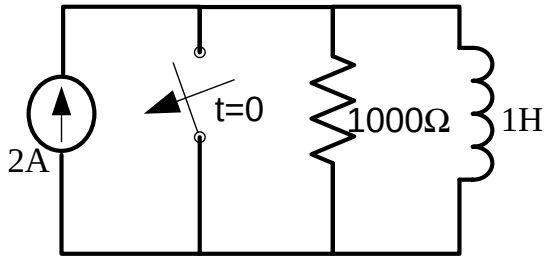
2A. Derive the signal $y(t) = 2x(-2t+3)+3$, from the signal shown below. Use appropriate signal transformation technique.



(04)

2B. The switch in the circuit shown is opened at t =0, Find v , $\frac{dv}{dt}$ and $\frac{d^2v}{dt^2}$ at t =0^+

Where 'v' is the voltage across the parallel branch.

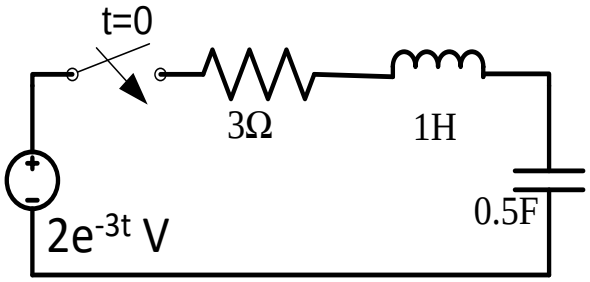


(03)

2C. A linear time invariant system is represented by a series RL circuit with R = 5Ω and L =2H. Determine the current response, if the circuit is excited by a voltage, **e(t) = 3u (t) + 5δ (t – 2).** Use time domain analysis technique.

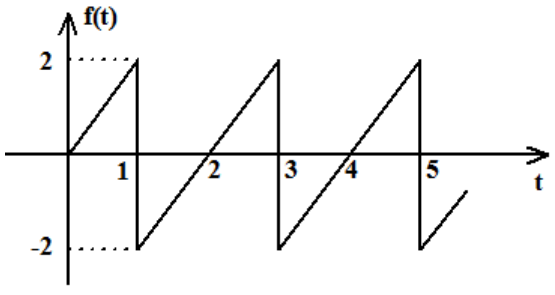
(03)

3A Obtain i(t), for the circuit shown below, using time domain approach.



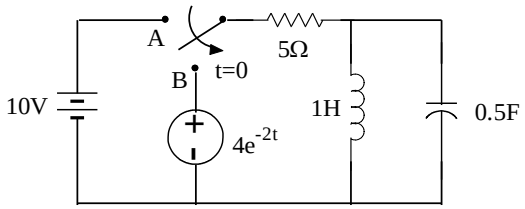
(04)

3B Find the Laplace transform of the periodic waveform shown below.



(02)

3C In the network shown, switch is changed from A to B at t = 0. Obtain S-domain network for t>0 (transformed network)



(04)

4A Obtain the exponential Fourier series representation and Fourier coefficients of the signal given below

$x(t) = 1 + sin\omega_o t + 2\cos\omega_o t + \cos(2\omega_o t + \frac{\pi}{4})$

Where ω_o is the fundamental frequency ***(03)***

4B Determine the Fourier series coefficients of the signal:

$x(t) = e^{-2t}$; $0 \le t \le 2$ repeating in every 2 units of time.

Plot the magnitude and phase spectrum for 7 frequency samples following symmetry and antisymmetric property, respectively. ***(04)***

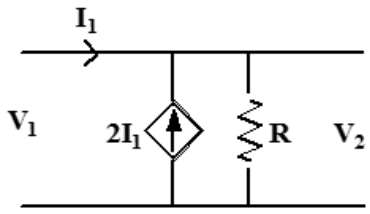
4C Obtain the magnitude spectrum of the signal $x(t)$, as a function of frequency using an appropriate Fourier tool. The signal $x(t)$ is described by the mathematical equation as below.

$$x(t) = \begin{cases} 1, & |t| < a \\ 0, & |t| > a \end{cases}$$

(03)

5A A resistive **π-** network is representative of a two-port model of an electronic circuit. The value of resistance is 2 Ω. Determine its open circuit parameters. ***(04)***

5B Derive the h-parameters of the two-port transformer circuit shown in figure below.



(02)

5C Two networks A and B are connected in parallel.

If $[Z_A] = \begin{bmatrix} 14 & 4 \\ 20 & 6 \end{bmatrix}$ and $[Y_B] = \begin{bmatrix} 1/s & -1/s \\ -1/s & 1/s \end{bmatrix}$,

Determine the Transmission (ABCD) parameters of the overall network.

(04)